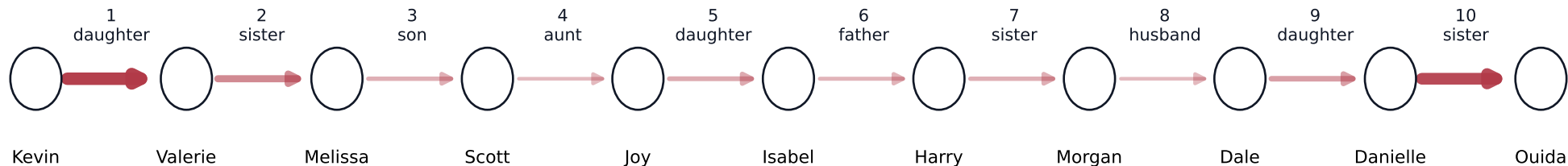


Long-Hop Failure Mode on a Real CLUTRR 10-Hop Example

Sample c357368f from data_089907f8 / 1.10_test.csv: query Kevin -> Ouida; target relation = granddaughter.

A. Shortcut-style attention over a 10-hop story

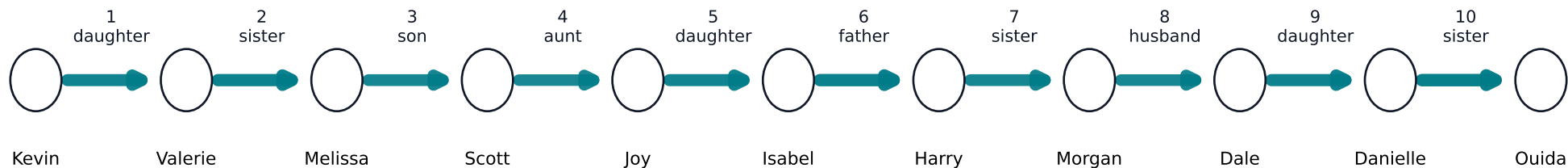
high mass on first/last edges, weak bridge



mid-chain evidence receives low mass; shortcut relies on endpoint/local kinship cues

B. TSRA trace-supervised reasoning path

step-by-step coverage of the full path



trace supervision keeps each intermediate relation active until the final composition

Composition: daughter -> sister -> son -> aunt -> daughter -> father -> sister -> husband -> daughter -> sister = granddaughter